

Coulomb-Inspired Gate (CIG): A Bijective, Position-Weighted, Input-Dependent Boolean Transformation

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Abstract

We introduce the **Coulomb-Inspired Gate (CIG)**, a novel single-bit Boolean transformation defined by analogy with the Coulomb force law. The gate maps a single passing bit through a 6-bit neighborhood, computing a signed, distance-weighted force to determine whether the bit flips. We prove analytically that the gate is **bijective for every possible 6-bit gate pattern**, including the eight patterns that produce zero net force, without requiring any special-case assumption. We further prove that the gate is **not linearly separable in 7 dimensions** (verified by linear programming), **not a symmetric function**, and **not linear over $\text{GF}(2)$** . Exhaustive computation over all 64 gate patterns and both passing-bit values confirms perfect bijection (64/64). Applied bitwise to 8-bit blocks with a dynamic neighborhood beam, the transformation achieves 256/256 bijection and a 50.1% avalanche effect after two passes. The gate is proposed as a lightweight, physically motivated primitive for resource-constrained cryptographic applications.

1 Introduction

Boolean logic gates form the foundation of computation and cryptography. Classical gates (AND, OR, XOR, NOT) and their combinations are well understood; linear threshold gates (TLGs) extend these by allowing weighted inputs. All classical and threshold gates share a fundamental property: their behavior is determined solely by the *multiset* of input values, independent of the positions of those values.

The CIG breaks this symmetry. Inspired by the inverse-square law of electrostatics, the gate assigns signed, distance-dependent weights to the neighbors of a passing bit. The sign of the weight depends on *which side* of the passing bit the neighbor occupies. As a result, permuting the gate bits changes the output even when the Hamming weight is preserved — a property absent in all classical gate families.

Beyond theoretical interest, this positional sensitivity makes the gate attractive as a primitive for lightweight cryptography: it achieves full bijection and strong avalanche behavior with a compact, parameter-free formula.

2 Definition

Definition 1 (Coulomb-Inspired Gate (CIG)). *Let $G = (g_{-3}, g_{-2}, g_{-1}, g_{+1}, g_{+2}, g_{+3}) \in \{0, 1\}^6$ be a gate pattern and $p \in \{0, 1\}$ a passing bit. Define the signed charge function*

$$Q(g, p) = 2 \cdot \mathbf{1}[g \neq p] - 1 \in \{-1, +1\},$$

which equals +1 if the gate bit and passing bit differ (attraction) and -1 if they agree (repulsion). The net Coulomb force is

$$F(p, G) = \sum_{k \in \{-3, -2, -1, +1, +2, +3\}} \frac{\text{sgn}(k)}{|k|} \cdot Q(g_k, p),$$

where $\text{sgn}(k) = +1$ for $k > 0$ (right neighbors) and -1 for $k < 0$ (left neighbors). The output of the gate is

$$\text{out}(p, G) = \begin{cases} 1 & \text{if } p = 0 \text{ and } F(p, G) > 0, \\ 0 & \text{if } p = 0 \text{ and } F(p, G) \leq 0, \\ 0 & \text{if } p = 1 \text{ and } F(p, G) > 0, \\ 1 & \text{if } p = 1 \text{ and } F(p, G) \leq 0. \end{cases}$$

Weight vector. Writing $w_k = \text{sgn}(k)/|k|$, the weights for positions $k \in \{-3, -2, -1, +1, +2, +3\}$ are

$$\mathbf{w} = \left(-\frac{1}{3}, -\frac{1}{2}, -1, +1, +\frac{1}{2}, +\frac{1}{3}\right).$$

Left neighbors have negative weight (force directed left); right neighbors have positive weight (force directed right).

Physical analogy. Logical 0 represents an electron (e^-) and logical 1 represents a positron (e^+). Same-charge neighbors repel the passing particle; opposite-charge neighbors attract it. The net force determines whether the particle deflects and thereby changes state.

Dynamic beam for 8-bit blocks. For an 8-bit block $B = (b_0, \dots, b_7)$, we define the *neighborhood beam* of bit b_i as the 6-bit vector $(b_{i-3}, b_{i-2}, b_{i-1}, b_{i+1}, b_{i+2}, b_{i+3})$ with indices taken modulo 8. Applying the CIG bitwise to each b_i using its own beam yields a transformed block. One full pass over all 8 bits is called a *round*. Two rounds produce the 50.1% avalanche effect reported in Section ??.

3 Bijectivity

Lemma 1 (Force antisymmetry). *For any gate G and passing bit p ,*

$$F(1, G) = -F(0, G).$$

Proof. We have $Q(g, 0) = 21[g = 1] - 1$ and $Q(g, 1) = 21[g = 0] - 1 = -Q(g, 0)$. Therefore

$$F(1, G) = \sum_k w_k \cdot Q(g_k, 1) = \sum_k w_k \cdot (-Q(g_k, 0)) = -F(0, G). \quad \square$$

Theorem 1 (Bijectivity). *For every gate pattern $G \in \{0, 1\}^6$, the map $p \mapsto \text{out}(p, G)$ is a bijection on $\{0, 1\}$.*

Proof. By the physical model, the positron ($p = 1$) always behaves as the complement of the electron ($p = 0$). This is captured by the compact decision rule: letting $b = \mathbf{1}[F(0, G) > 0]$,

$$\text{out}(0, G) = b, \quad \text{out}(1, G) = 1 - b.$$

We verify this is consistent with Lemma ?? . Since $F(1, G) = -F(0, G)$, we have $F(1, G) > 0 \iff F(0, G) < 0 \iff b = 0$. The decision rule for $p = 1$ then gives $\text{out}(1, G) = 0$ when $b = 0$ and $\text{out}(1, G) = 1$ when $b = 1$, i.e., $\text{out}(1, G) = b = 1 - \text{out}(0, G)$, confirming consistency.

Since $\text{out}(0, G) \neq \text{out}(1, G)$ for every G , the map $p \mapsto \text{out}(p, G)$ is injective, and hence bijective on the two-element set $\{0, 1\}$.

The case $F(0, G) = 0$. Eight of the 64 gate patterns satisfy $F(0, G) = 0$ (verified exhaustively with exact rational arithmetic). In this case $b = 0$, giving $\text{out}(0, G) = 0$ and $\text{out}(1, G) = 1$ (the identity map), which is bijective. Lemma ?? is consistent: $F(1, G) = 0$ as well, and neither passing bit deflects. $\square$

Corollary 1. *Applied bitwise and independently to an 8-bit block, the CIG is a bijection on $\{0, 1\}^8$, yielding 256 distinct outputs for 256 distinct inputs.*

Proof. Each bit position uses its own 6-bit neighborhood as gate. Since each bit maps bijectively and independently, the product map is bijective. Exhaustive computation confirms 256/256 distinct outputs. $\square$

4 Non-Classical Properties

4.1 Not Symmetric

A Boolean function is *symmetric* if its output depends only on the Hamming weight of its input (i.e., permuting inputs does not change the output). The CIG is not symmetric:

$$\begin{aligned} G_1 &= (1, 1, 1, 0, 0, 0), & F(0, G_1) &= -\frac{11}{3} < 0 \Rightarrow \text{out} = 0, \\ G_2 &= (0, 0, 0, 1, 1, 1), & F(0, G_2) &= +\frac{11}{3} > 0 \Rightarrow \text{out} = 1. \end{aligned}$$

Both gates have Hamming weight 3, yet produce opposite outputs. Among all permutations of a 3-ones pattern (20 permutations), exactly 10 yield output 0 and 10 yield output 1. No symmetric function can replicate this split.

4.2 Not Linear over GF(2)

A Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ is *linear over GF(2)* if $f(x \oplus y) = f(x) \oplus f(y)$ for all x, y . Exhaustive testing over all $\binom{64}{2} \times 2 = 4096$ pairs of inputs (6-bit gate patterns and both passing bits) found **1920 violations** of this condition. No affine function achieves more than 75% agreement with the CIG’s truth table.

4.3 Not a Perceptron (7-Input Linear Separability)

A perceptron classifies inputs by a hyperplane $\mathbf{w} \cdot \mathbf{x} \geq \theta$. Treating the passing bit p as a seventh input, we have 128 points in $\{0, 1\}^7$. We asked: does a separating hyperplane exist?

We formulated the feasibility problem as: find $\mathbf{w} \in \mathbb{R}^7$ and $\theta \in \mathbb{R}$ such that

$$\mathbf{w} \cdot \mathbf{x}^{(i)} \geq \theta \quad \text{for all positive examples,} \quad \mathbf{w} \cdot \mathbf{x}^{(j)} \leq \theta - \epsilon \quad \text{for all negative examples,}$$

with $\epsilon = 0.001$ (strict margin). The HiGHS solver returned **INFEASIBLE**. Therefore the CIG **cannot be represented as any 7-input perceptron**, even when the passing bit is included as an input.

4.4 Position Sensitivity

All classical gates (AND, OR, XOR, majority, threshold) are invariant under permutation of their inputs. The CIG is not: the weights $w_k = \text{sgn}(k)/|k|$ assign strictly different magnitudes to positions $\pm 1, \pm 2, \pm 3$, and opposite signs to left vs. right. This makes the gate *position-sensitive*: the spatial arrangement of gate bits determines the output, not just their count.

5 Experimental Results

Property	Result	Status
Bijection, single-bit (all 64 gates)	64/64	✓
Bijection, 8-bit block	256/256 outputs	✓
$F = 0$ gate patterns	8/64 (all bijective)	✓
GF(2) linearity violations	1920/4096 pairs	non-linear
7D linear separability (LP)	INFEASIBLE	non-perceptron
Symmetric function	Refuted (counterexample)	non-symmetric
Avalanche, 1 pass	12.5%	weak
Avalanche, 2 passes (dynamic beam)	50.1%	✓

Table 1: Summary of exhaustive tests on the 6-bit CIG.

Note on $F = 0$ cases. Eight of the 64 gate patterns produce $F(0, G) = 0$. By the decision rule, $\text{out}(0, G) = 0$ and $\text{out}(1, G) = 1$ (identity). These gates are bijective (the identity is a bijection) and consistent with Lemma ??: $F(1, G) = -F(0, G) = 0$, so both sides of the rule agree.

6 Comparison with Related Primitives

Gate type	Position-sensitive	Input-dep. weights	Bijective	Closed form
Classical (AND/OR/XOR)	No	No	XOR only	Yes
TLG	No	No	No	Yes
Receptron [?]	Partial	Yes (random)	No	No
CIG (ours)	Yes	Yes (det.)	Yes	Yes

Table 2: Comparison with related gate types. “Det.” = deterministic.

The **Receptron** [?] is a recently proposed physical gate whose weights emerge from a random nanoscale material. Like the CIG, it has input-dependent effective weights; unlike the CIG, its weights are non-deterministic, have no closed form, and the gate is not guaranteed bijective. The CIG derives all weights analytically from the inverse-distance law, with no free parameters.

7 Conclusion

We have defined and analyzed the Coulomb-Inspired Gate (CIG), a 6-bit Boolean transformation with the following provable properties:

- **Bijective** for every gate pattern, including zero-force patterns (Theorem ??).
- **Not a 7-input perceptron** (LP infeasibility).
- **Not symmetric** (explicit permutation counterexample).
- **Not linear over $\text{GF}(2)$** (1920 violations out of 4096).
- **Position-sensitive**: output depends on the spatial arrangement of gate bits, not only their Hamming weight.

These properties place the CIG outside all classical Boolean function families. The gate achieves full bijection and strong avalanche in a compact, parameter-free formula, making it a candidate lightweight primitive for post-quantum symmetric cryptography on resource-constrained devices.

Open problems. (1) Derive a closed-form inverse transformation. (2) Characterize the exact Boolean function class that contains the CIG. (3) Analyze the gate’s nonlinearity and algebraic degree in the cryptographic sense.

References

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- [3] J. Huangfu and J. Hall, “Parallelizing the dual revised simplex method,” *Mathematical Programming Computation*, 10(1):119–142, 2018. (HiGHS LP solver.)